

Parva Shah

Data Engineer

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SUMMARY

Experienced Software Developer with proven expertise in improving system efficiency and delivering impactful results. Enhanced data processing by 40% using PySpark, optimized test execution time by 60% with Python automation, and optimized AWS IoT pipelines, increasing dataset accuracy by 75%. Launched scalable RESTful APIs and generated real-time analytics tools, including a Stock Market Simulator, leveraging APIs and Snowflake. Published deployment tools at Amazon, boosting operational efficiency by 20%. Holds a Master's in Computer Science from the University of Texas at Arlington.

PROFESSIONAL EXPERIENCE

Python/Data Engineer | FANNIE MAE, USA

Feb 2023 – Present

- Provisioned and managed AWS infrastructure using Terraform, improving resource allocation efficiency by 30%.
- Implemented and optimized ETL pipelines with PySpark, enabling faster data transformation and reducing processing times by 40%.
- Transformed REST API endpoints using Flask and Django, ensuring seamless integration and scalability, supporting a 25% increase in data throughput.
- Automated CI/CD pipelines with Jenkins and Bitbucket, reducing deployment cycles by 20% while maintaining robust quality assurance.
- Conducted comprehensive unit testing with Pytest, ensuring a 95% code accuracy rate and minimizing bugs in production.
- Enhanced incident response by automating log analysis and alerts with AWS CloudWatch, reducing resolution time by 50%.

Software Development Engineer | AMAZON, USA

Aug 2021 – Feb 2023

- Enhanced the Dafny compiler and interpreter, increasing functionality and performance by 30% through efficient C# implementations.
- Reduced automated test execution time by 60% through GraphQL API optimizations and streamlined test scripts.
- Automated ETL processes with Python, decreasing manual workload by 70% and improving data consistency.
- Prepared Python and Shell scripts to monitor nightly releases against production builds, ensuring 100% reliability in deployment.
- Built tools using CloudWatch and DynamoDB insights, improving server performance monitoring and usage tracking accuracy by 25%.
- Published internal deployment tools, increasing adoption by 20% and reducing installation overhead for engineering teams.

Network/IoT Development Intern | DARKHORSE, USA

Sep 2020 – Jan 2021

- Programmed Nordic Semiconductor devices to record and transmit accelerometer and GPS data, achieving a 30% reduction in data transfer times using LoRaWAN protocols.
- Created AWS Lambda pipelines, optimizing real-time data transfer efficiency and accuracy.
- Improved measurement accuracy for player speed and directional analysis by 75% using DynamoDB datasets and custom optimization algorithms.

Web Development Intern | ABHINAAM CHESS ACADEMY, INDIA

Jan 2019 – Jun 2019

- Designed and deployed a website for managing students, employees, and events, enhancing administrative efficiency by 40%.
- Developed a gamified point system for students using Django and Node.js, increasing student engagement by 25%.
- Integrated Stockfish-based chess puzzles into the platform, boosting learning outcomes for chess students.

EDUCATION

- MS in Computer Science, **University of Texas at Arlington, USA**
- BE in Computer Engineering, **Gujarat Technological University, India**

Aug 2019 to May 2021

Aug 2015 to May 2019

TECHNICAL SKILLS

Category	Skills
Programming Languages	Python (Pandas, NumPy, Matplotlib, Seaborn, Scikit-learn, TensorFlow, Keras, OpenCV), SQL, Java, JavaScript, C, C++
Reporting & BI Tools	Power BI, Tableau, Microsoft Excel, AWS Quicksight, Jupyter Notebook, Git, WaveFront, Databricks, JIRA, Concourse, Swagger
Databases	MySQL, MS SQL Server, PostgreSQL, Oracle, MongoDB
Machine Learning & AI	Regression (Linear, Logistic, Bayesian, Ridge, Lasso), Classification (Decision Trees, Random Forest, SVM, Naive Bayes, k-Nearest Neighbors, Neural Networks), Clustering (K-means, Hierarchical), Anomaly Detection, XGBoost, LSTM, RNN, Time Series Forecasting
Cloud & Big Data	Google Cloud Platform (BigQuery, Cloud Composer, Dataflow, Data Proc, Compute Engine, Google Kubernetes Engine, App Engine, GCP Storage Buckets), AWS (S3, Redshift, EC2, Glue), Azure, Pivotal Cloud Foundry, Hadoop, Apache Spark, Data Lake Management
ETL & Data Pipelines	Apache Kafka, Apache NiFi, Hive, Airflow, Ansible, Telegraph, Data Pipeline Automation, Building Batch and Streaming Pipelines
Natural Language Processing	NLP techniques, Sentiment Analysis, Text Processing with NLTK and TensorFlow, Speech Recognition, Social Media Analytics, and Brand Perception Analysis
Frameworks & Libraries	Flask, Django, Falcon, PySpark, Prophet
Version Control & Collaboration	Git, Source Tree, Vowpal Wabbit, TensorBoard
Operating Systems	Linux, Unix, Windows, MacOS